

① 空间嵌入

Input
 $B \times 32 \times 26 \times 3$

1×1 Conv
 $3 \rightarrow 64$ | LayerNorm
Joint Pool (J:26 $\rightarrow$ 1) + PosEnc

$B \times 32 \times 64$

② 多尺度空洞卷积

Conv1d $k=3, d=1, 64ch$
BN + GELU

Conv1d $k=3, d=2, 64ch$
BN + GELU

Conv1d $k=3, d=4, 64ch$
BN + GELU

③ 自注意力 4 heads

Linear $64 \rightarrow 64 \times 3$ (Q, K, V)

4-Head Split $4 \times (T \times 16)$

softmax $(QK^T / \sqrt{16}) \cdot V$

Concat + Project

+ Residual + LN + Dropout(0.1)

④ 分类头

GlobalAvgPool
 $T\text{-dim} \rightarrow 1, 64d$

FC $64 \rightarrow 32$ + GELU

FC $32 \rightarrow 7$

Softmax
 $p \in \mathbb{R}^7$
 $g = \text{argmax } p \mid s = \text{max } p$

57K 参数 (CNN-only: 40K) | **221 KB** float32 | **1.32 ms** Snapdragon XR2 (Unity Barracuda)